

Chapter 9: Products and Cayley Graphs

9.1 Product Graphs

There are multiple ways to combine two graphs to get a new one. For a product of graphs G and H , the vertex set is the set of ordered pairs (g, h) where $g \in V(G)$ and $h \in V(H)$. The most common product is the **cartesian product** written $G \square H$. Here two pairs are adjacent if they are the same in one coordinate and adjacent in the other. For example: hypercube $Q_k = Q_{k-1} \square K_2$. The cartesian product has a spanning subgraph consisting of copies of G and one consisting of copies of H ; these copies are called its **fibers**. The most famous result about cartesian products is the formula for their chromatic number: $\chi(G \square H) = \max(\chi(G), \chi(H))$.

The product $G \times H$ has several names including **direct product** and **tensor product**. (Note that some people use notation $\times$ for cartesian product.) In the direct product two pairs are adjacent if both their first and second coordinates are adjacent. Note for example that if both G and H are connected nontrivial bipartite graphs, then $G \times H$ is bipartite but has two components. And in general $\chi(G \times H) \leq \min(\chi(G), \chi(H))$. For 50 years it was conjectured that $\chi(G \times H) = \min(\chi(G), \chi(H))$, but this was disproved by Shitov.

Another product $G \boxtimes H$ also has several names including **strong produce** and **compositioin**. Here two pairs are adjacent if their first coordinates are adjacent, or if they are the same in the first coordinate and adjacent in the second. Essentially one takes G and replaces each vertex by a copy of H ; the resultant edges are the union of the cartesian and direct products.

9.2 Cayley Graphs

Let Γ be a group and S a subset of Γ . Then the **Cayley graph** generated by S has vertex set Γ ; and an arc (α, β) is present if and only if $\beta = \alpha\sigma$ for some $\sigma \in S$. That is, $\alpha^{-1}\beta \in S$. For example, if we take $\mathbb{Z}_n$ with set $S = \{1\}$, then the resultant Cayley graph is just the (directed) n -cycle. We assume the identity is not in S , so there are no loops. If S generates Γ , then the Cayley graph is

strongly connected. If S is closed under inverses, then one can consider the result an undirected graph. Note that every Cayley graph is regular.

The hypercube is an example of a Cayley graph. Indeed, if G and H are Cayley graphs, then their Cartesian product is also a Cayley graph, namely of the direct product of the groups for G and H .

Cayley graphs have an immediate symmetry. A graph is called **vertex-transitive** if all vertices are identical. That is, for every pair of vertices u and v there is an automorphism of the graph that maps u to v .

THEOREM. *A Cayley graph is vertex-transitive. Indeed, for any $\lambda \in \Gamma$, the mapping $\alpha \rightarrow \lambda\alpha$ is an automorphism.*

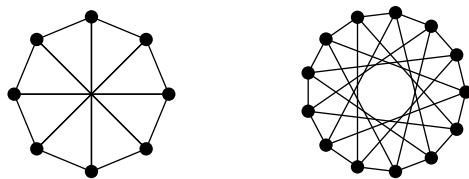
PROOF. We know (α, β) is an arc if and only if $\alpha^{-1}\beta \in S$. Thus, $(\lambda\alpha, \lambda\beta)$ is an arc if and only if $(\lambda\alpha)^{-1}(\lambda\beta) \in S$. But $(\lambda\alpha)^{-1}(\lambda\beta) = \alpha^{-1}\beta$. $\square$

In fact, it follows that the automorphism group of the Cayley graph has a subgroup isomorphic to Γ . An undirected Cayley graph also has connectivity properties: for example if it is connected then its edge-connectivity equals its degree.

A nonexample is the Petersen graph. Though the Petersen graph is vertex-transitive, it is not a Cayley graph. The Petersen graph is a special case of the **Kneser graphs**, most of which are not Cayley. A Kneser graph is defined by taking all k -element subsets of an n -element universe, and making two subsets adjacent if and only if they are disjoint. The complement of a Kneser graph is a **Johnson graph**. An example of a Johnson graph is the line graph $L(K_m)$, which for even $m \geq 6$, is not Cayley.

9.3 Circulants and Paley Graphs

A **circulant graph** is where Γ is the cyclic group. Note that both K_n and $K_{m,m}$ are circulants. We can denote an undirected circulant by $C_n[a_1, \dots, a_r]$ where $S = \{\pm a_1, \dots, \pm a_r\}$. For example, here are the circulants $C_8[1, 4]$ (also called a Möbius ladder) and $C_{13}[1, 5]$.



A famous circulant is the ***Paley graph***. Let p be a prime power such that $p \equiv 1 \pmod{4}$. (Note that for such p it follows that -1 has a square-root.) Then define the circulant on p vertices with S the set of quadratic residues; that is, $S = \{a^2 \bmod p : 1 \leq a \leq p-1\}$. For example, the Paley graph on 5 vertices is the 5-cycle, while the Paley graph on 13 vertices is $C_{13}[1, 3, 4]$. One can show that the Paley graph is self-complementary (isomorphic to its complement). (If one takes a prime power such that $p \equiv 3 \pmod{4}$ then one gets a Paley digraph.)

Paley graphs are famous as examples of ***quasi-random graphs***: that is, explicit constructions that have “all” the properties of random graphs. In particular they are what are called ***expander*** graphs.